

Supplement: Real-World Hard-case VQA Pool for Agentic Multistage Checkpoint Evaluation

TLDR

This 100-sample hard-case VQA pool demonstrates when multistage agentic checkpoint evaluation is informative and when it is not. As image quality and OCR readability deteriorate, pointwise scores become increasingly compressed across checkpoints, reducing checkpoint separability and leading to higher disagreement or ties in subsequent listwise and pairwise stages. Conversely, on the approximately 20% of samples with sufficient visual evidence, the three-stage pipeline produces grounded and interpretable checkpoint preferences. These results show that evaluation confidence should be conditioned on input evaluability, making multistage checkpoint selection quality-aware rather than universally applied.

Purpose: Support material explaining why this sample pool is presented, how it validates the proposed multistage agentic evaluation method, and where its strengths and limitations lie.

S1. Motivation and Overview

We evaluate checkpoint selection on a 100-sample hard-case egocentric VQA pool drawn from a DPO training readout. Unlike curated academic benchmarks, this dataset intentionally reflects realistic deployment conditions, including blurred images, glare, small text, OCR-heavy questions, and heterogeneous visual tasks. The objective is not simply to identify the best checkpoint, but to understand under what conditions multistage agentic evaluation produces reliable checkpoint preferences.

The central observation is that evaluation quality is fundamentally constrained by input quality. As image quality and OCR readability deteriorate, visual evidence becomes insufficient for reliable judgment, causing Stage I pointwise scores to become increasingly compressed across checkpoints. Reduced score separation then propagates through the evaluation pipeline, leading to higher disagreement and tie rates during the subsequent listwise and pairwise stages. Conversely, when sufficient visual evidence exists (approximately 20% of the pool), Stage I clearly separates checkpoints and the later stages consistently refine those preferences through complementary ranking strategies.

Rather than treating all evaluation samples equally, this supplement demonstrates that multistage checkpoint evaluation should be interpreted together with input evaluability. The proposed quality-aware analysis therefore distinguishes between uncertainty caused by weak checkpoints and uncertainty caused by insufficient visual evidence, providing a more reliable basis for checkpoint selection under real-world deployment conditions.

This pool is **not a curated academic benchmark**. It is intentionally **real-world**:

- Egocentric photos with blur, glare, and small text
- Code-mixed questions
- Heterogeneous tasks (OCR, scene description, UI help, refusal)

We include it because production VQA evaluation faces exactly this noise. A method that only works on clean laboratory data is insufficient.

This supplement demonstrates:

1. The multistage pipeline runs end-to-end on realistic inputs.
2. Stratified analysis reveals when checkpoint rankings are trustworthy.
3. Input quality (especially OCR) is a first-order determinant of evaluation validity rather than a minor caveat.

S2. Experimental Setup

Item	Value
Samples (N)	100
Checkpoints (C)	7 (dpo_enhin_I_2000 – _20000)
Ground truth	None (judges see image + question + response only)
Stage I	Pointwise scoring (Claude Sonnet 4.6), 700 judgments
Stage II	Listwise ranking, top-6 after Stage I screen (Borda)
Stage III	Pairwise comparison, finalists 16000 vs. 20000 (GPT-4.1)
Bootstrap	200 rounds (Stage I–II stability)
Pre-funnel quality pass	Evaluability on image + question only (no checkpoint responses)

S3. Pool Composition and Quality Strata

Table 1 – Evaluability Buckets

Quality bucket	Count	Share	Interpretation
Low	84	84%	Required evidence not readable or question not answerable from image
Medium	4	4%	Partial evidence
High	12	12%	Clear visual evidence for evaluation

Table 2 – OCR / Task Strata

Samples are tagged using a pre-funnel OCR and task analysis (evaluability.csv, low_quality_failure_analysis.csv).

[†] Mean of visual evaluability, evidence readability, and answerability.

[‡] Stage I aggregated over seven checkpoints per sample.

[§] Non-OCR or OCR-optional tasks where grounded evaluation is possible (scene/object identification, QR scan, proper refusal).

Stratum	N	Share	Evaluability [†]	Stage I Mean [‡]	Halluc. Penalty [‡]	Stage III Tie Rate
<code>ocr_better</code>	9	9%	0.67	0.71	0.16	78%
<code>method_valid</code> [§]	11	11%	0.45	0.67	0.19	55%
<code>ocr_mid</code>	6	6%	0.17	0.52	0.24	67%
<code>ocr_unreadable</code>	49	49%	0.00	0.32	0.33	69%
<code>other</code>	25	25%	0.08	0.30	0.40	48%

Key observation. Checkpoint separability closely follows visual evaluability. Nearly half of the pool (49%) requires OCR but contains unreadable text, causing Stage I scores to collapse across checkpoints. Only the `method_valid` and `ocr_better` strata (approximately 20% of the pool) contain sufficient evidence for meaningful checkpoint discrimination.

S4. How This Pool Supports the Multistage Method

Table 3 – Overall Funnel Results (100 Samples)

Stage	Top checkpoint	Stability (bootstrap)	Outcome
I – Pointwise	8000	0.31 consistency (0.69 flip)	Soft screen; means clustered (~ 0.38 – 0.41)
II – Listwise	16000	0.50 consistency	Ranks six survivors; 16000 = 20000
III – Pairwise	–	–	$\sim 63\%$ ties; 16% ~ 16000 , 21% ~ 20000

The disagreement across stages is therefore expected rather than contradictory. When Stage I cannot reliably distinguish checkpoints because of limited visual evidence, Stage II and Stage III naturally exhibit unstable rankings and high tie rates. In contrast, on evaluable samples, the later stages primarily refine rather than recover checkpoint preferences.

Table 4 – Where Multistage Evaluation is Informative

Evaluation Tier	Criteria	N	Role
Tier 1	Evaluability ≥ 0.5 and Stage I mean ≥ 0.5	7	Highest-confidence demonstrations with clear judge grounding.
Tier 2	Primary <code>method_valid</code> + <code>ocr_better</code>	20	Decision-grade subset for checkpoint comparison.
Tier 3	<code>method_valid</code>	21	Representative examples illustrating Stages I–III.
Out of scope	<code>ocr_unreadable</code> + <code>other</code>	74	Report as limitations; exclude from checkpoint ranking.

On **Tier 2** (20 samples), the mean Stage I score is 0.69. Checkpoints separate on transcription quality (`ocr_better`) and visual grounding (`method_valid`). Multistage disagreement therefore reflects genuine preference aggregation (mean vs. rank vs. head-to-head), rather than input degeneracy.

On OCR-blocked strata, all stages still execute, but Stage I assigns uniformly low scores (8 of 49 unreadable samples exhibit zero spread across checkpoints), while Stages II–III occasionally

prefer verbose but ungrounded responses. This is a failure mode exposed by the stratified design rather than evidence of successful multistage ranking.

What the Method Demonstrates

Strength	Evidence from this pool
Scalable pipeline	100 $\times$ 7 pointwise evaluations followed by listwise and pairwise refinement completed in a single evaluation session.
Progressive refinement	Stage I filters weak checkpoints, while Stage II and Stage III progressively refine comparisons only where sufficient evidence exists.
Honest uncertainty	High pairwise tie rates (63%) and low bootstrap consistency explicitly quantify evaluation uncertainty rather than forcing unstable rankings.
Quality-aware evaluation	Pre-funnel evaluability and OCR analysis predict checkpoint separability and indicate when multistage rankings are reliable.
Real-world coverage	Egocentric VQA with OCR-heavy and heterogeneous tasks better reflects deployment conditions than clean academic benchmarks.

S5. Strengths and Limitations

Strengths

1. **Realistic evaluation.** The pool reflects production VQA where poor image quality is often a larger limitation than model capability.
2. **Quality-aware checkpoint selection.** The evaluation explicitly identifies when checkpoint rankings are trustworthy instead of assuming all samples contribute equally.
3. **Diagnostic insight.** The stratified analysis separates model limitations from input-quality limitations, avoiding misleading interpretations of checkpoint convergence.
4. **Transparent uncertainty.** Bootstrap instability and pairwise tie rates quantify confidence in checkpoint selection rather than forcing a single ranking.

Limitations

1. Only a small fraction (7–20 samples) provides sufficiently strong evidence for confident checkpoint comparison.
2. As OCR readability decreases, Stage I scores become increasingly compressed across checkpoints ($0.71 \rightarrow 0.52 \rightarrow 0.32$), reducing checkpoint separability.
3. When Stage I cannot distinguish checkpoints, downstream rankings may reflect response fluency rather than grounded visual correctness.

4. Even on readable OCR samples, high pairwise tie rates indicate that multistage evaluation appropriately reports checkpoint equivalence rather than over-interpreting negligible differences.
5. No external ground-truth annotations are available; evaluation relies on grounded visual judgment together with the independent evaluability analysis.

S6. Recommended Reporting Practice

When citing this pool:

1. Report strata separately (Table 2). Never combine OCR-blocked and evaluable samples when making checkpoint selection claims.
2. Present Tier 2 (20 samples) as the primary quantitative evidence and Tier 1 (7 samples) for qualitative examples.
3. Treat full-pool aggregates (63% ties and stage disagreement) as limitation analysis rather than the recommended operating point.
4. Explicitly state the dependency chain:

Image OCR Quality → Evaluability → Judge Grounding → Checkpoint Separability → Stage Agreement

5. Pair multistage evaluation with an explicit quality gate. Samples failing evaluability or OCR thresholds should be reported as low-confidence.